ITCS3166
Computer Networks

Lecture 2 Physical Layer

Dr. Dazhao Cheng
Department of Computer Science

Example

- An image is 640 X 480 pixels with 2 bytes/pixel. Assume the image is uncompressed. How long does it take to transmit it over a 56-kbps modem channel? Over a 1.5 Mbps DSL?

Example

- **Imagine that you have trained your dog to carry a box of three 8mm tapes instead of a flask of brandy. These tapes each contain 7 GBs. The dog can travel at a rate of 18 km/hr. For what range of distances does Bernie have a higher data rate than 150 Mbps?**

Ans: The dog can carry 21 gigabytes, or 168 gigabits. A speed of 18 km/hour equals 0.005 km/sec. The time to travel distance x km is x / 0.005 = 200x sec, yielding a data rate of 168/200x 2^30 bps or 901.9431322/x Mbps if G is defined as **2^30**.

For x < 6.01 km, the dog has a higher rate than the communication line.

But G is defined as **10^9** in most commercial products. Then the data rate is 168/200x 10^9 bps or 840/x Mbps.

For x < 5.6 km, the dog has a higher rate than the communication line.

Questions of Interests

- How long will it take to transmit a message?
  - How many bits are in the message (text, image)?
  - How fast does the network/system transfer information?
- Can a network/system handle a voice (video) call?
  - How many bits/second does voice/video require? At what quality?
- How long will it take to transmit a message without errors?
  - How are errors introduced?
  - How are errors detected and corrected?
- What transmission speed is possible over radio, copper cables, fiber, …?

A Transmission System

Transmitter

- Converts information into *signal* suitable for transmission
- Injects energy into communications medium or channel
  - Telephone converts voice into electric current
  - Modem converts bits into tones

Receiver

- Receives energy from medium
- Converts received signal into form suitable for delivery to user
  - Telephone converts current into voice
  - Modem converts tones into bits

Transmission Impairments

Communication Channel

- Pair of copper wires
- Coaxial cable
- Radio
- Light in optical fiber
- Light in air
- Infrared

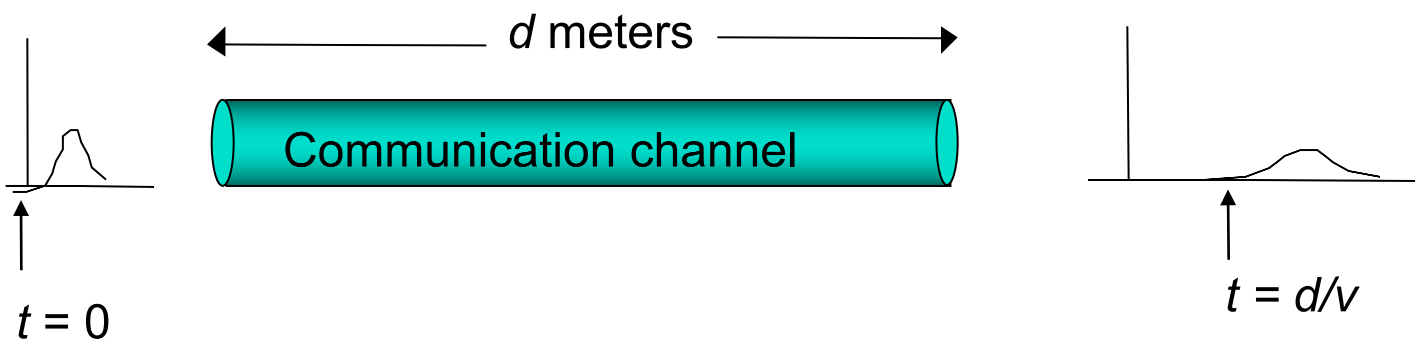
Transmission Impairments

- Signal attenuation
- Signal distortion
- Spurious noise
- Interference from other signals

Bit Rates of Digital Transmission Systems

System	Bit Rate	Observations
Telephone twisted pair	33.6-56 kbps	4 kHz telephone channel
Ethernet twisted pair	10 Mbps, 100 Mbps	100 meters of unshielded twisted copper wire pair
Cable modem	500 kbps-4 Mbps	Shared CATV return channel
ADSL twisted pair	64-640 kbps in, 1.536-6.144 Mbps out	Coexists with analog telephone signal
2.4 GHz radio	2-11 Mbps	IEEE 802.11 wireless LAN
28 GHz radio	1.5-45 Mbps	5 km multipoint radio
Optical fiber	2.5-10 Gbps	1 wavelength
Optical fiber	>1600 Gbps	Many wavelengths

Fundamental Issues in Transmission Media



° **Propagation speed of signal**

- **c = 3 x 10^8 meters/second in vacuum**
- **$\nu = c/\sqrt{\varepsilon}$ speed of light in medium where $\varepsilon > 1$ is the dielectric constant of the medium**
- **ν = 2.3 x 10^8 m/sec in copper wire; ν = 2.0 x 10^8 m/sec in optical fiber**

The Theoretical Basis for Data Communication

Data information can be transmitted on wires by varying some physical property as voltage.

- Fourier Analysis
- Bandwidth-Limited Signals
- Maximum Data Rate of a Channel

Guided Transmission Data

What is the purpose of the physical layer?

- Magnetic Media
- Twisted Pair
- Coaxial Cable
- Fiber Optics

Magnetic Media

- Given a 200GB tape, a box 60 X 60 X 60cm can hold about 1000 of these tapes. A box can be delivered anywhere domestically in 24 hours. What is the effective bandwidth?

What are the major problems with magnetic media?

Twisted Pair

- A twisted pair consists of two insulated copper wires, typically about 1mm thick.
  - More twists per cm leads to less crosstalk and better quality over longer distance

(a)

Why twisted pairs are widely used?

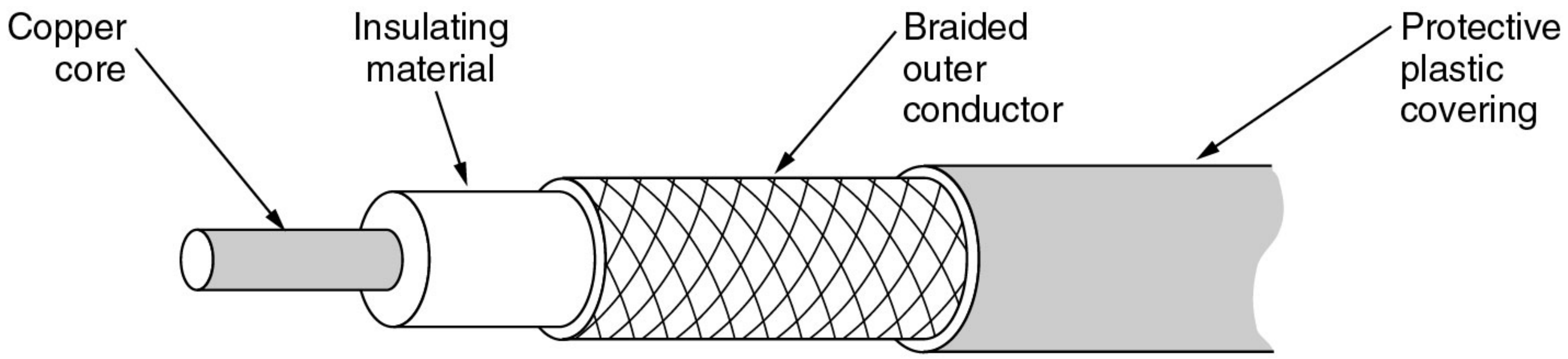
(b)

(a) Category 3 UTP (16 MHz).

(b) Category 5 UTP (100 MHz).

Coaxial Cable

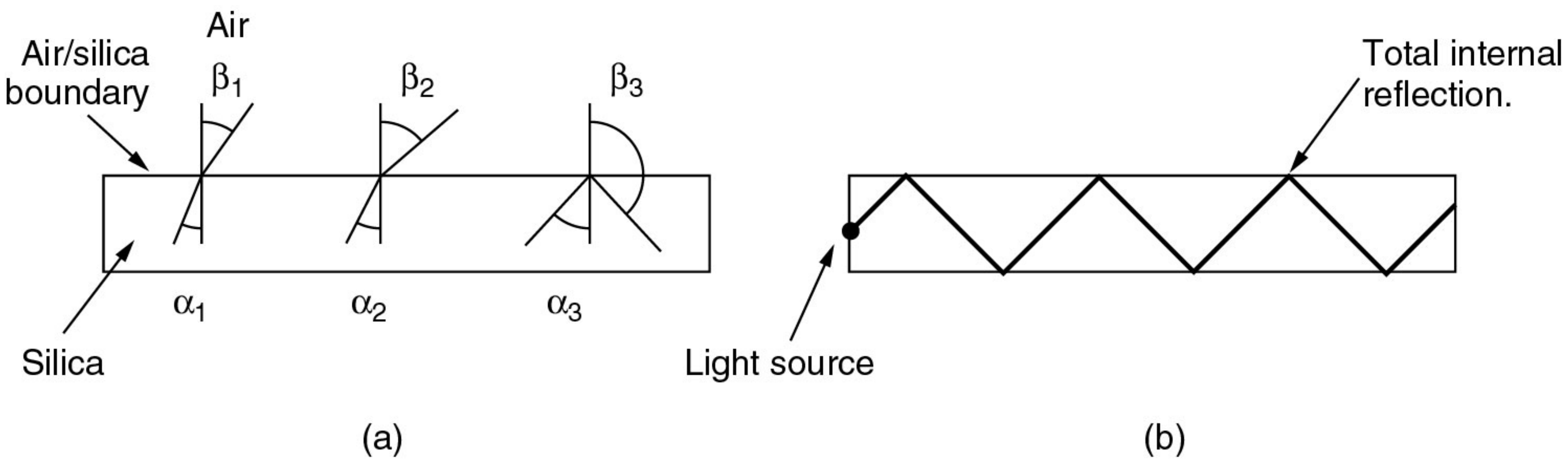
- A good combination of high bandwidth and excellent noise immunity.
  - Used to be in telecom for long-distance lines (replaced by fiber optics), but still widely used in cable TV and MANs.



A coaxial cable.

Fiber Optics

- An optical transmission system has three key components: the light source, transmission medium, and the detector
  - Main issue: light leaking



(a) Three examples of a light ray from inside a silica fiber impinging on the air/silica boundary at different angles.

(b) Light trapped by total internal reflection (multimode fiber).

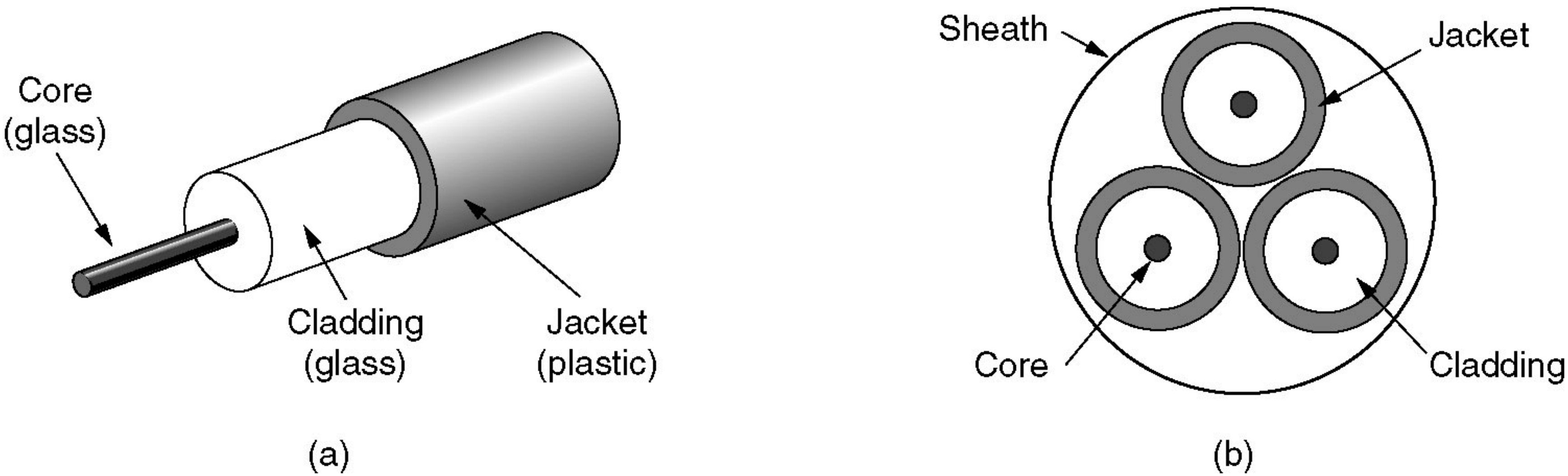
The achievable bandwidth is more than 50 Tbps. Why the current practical singling limit is about 10Gbps (100 Gbps in Labs)?

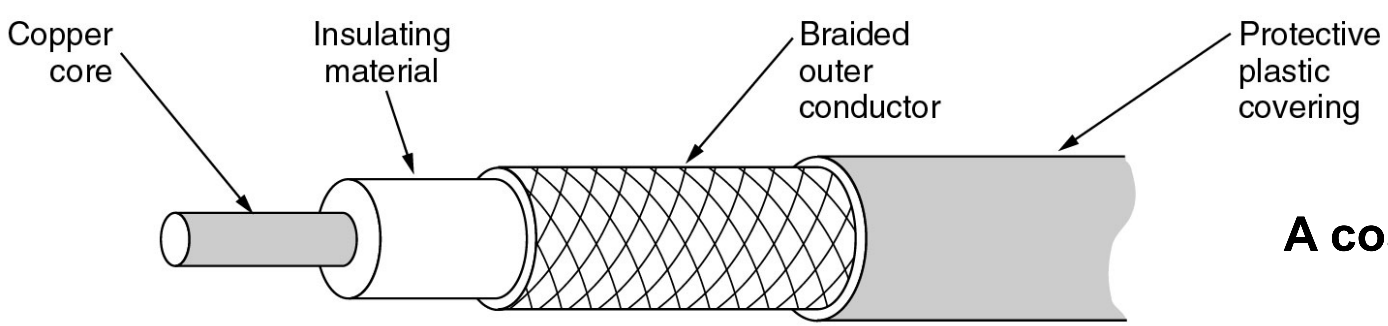
Who wins the rate between computing and communication?

Fiber Cables

(a) Side view of a single fiber.

(b) End view of a sheath with three fibers.





A coaxial cable.

Questions?

- **What are the advantages of fiber optics over copper as a transmission medium? Is there any downside of using fiber optics over copper?**

- **Fiber has many advantages over copper. It can handle much higher bandwidth than copper. It is not affected by power surges, electromagnetic interference, power failures, or corrosive chemicals in the air. It does not leak light and is quite difficult to tap. Finally, it is thin and lightweight, resulting in much lower installation costs.**

- **There are some downsides of using fiber over copper. First, it can be damaged easily by being bent too much. Second, optical communication is unidirectional, thus requiring either two fibers or two frequency bands on one fiber for two-way communication. Finally, fiber interfaces cost more than electrical interfaces.**

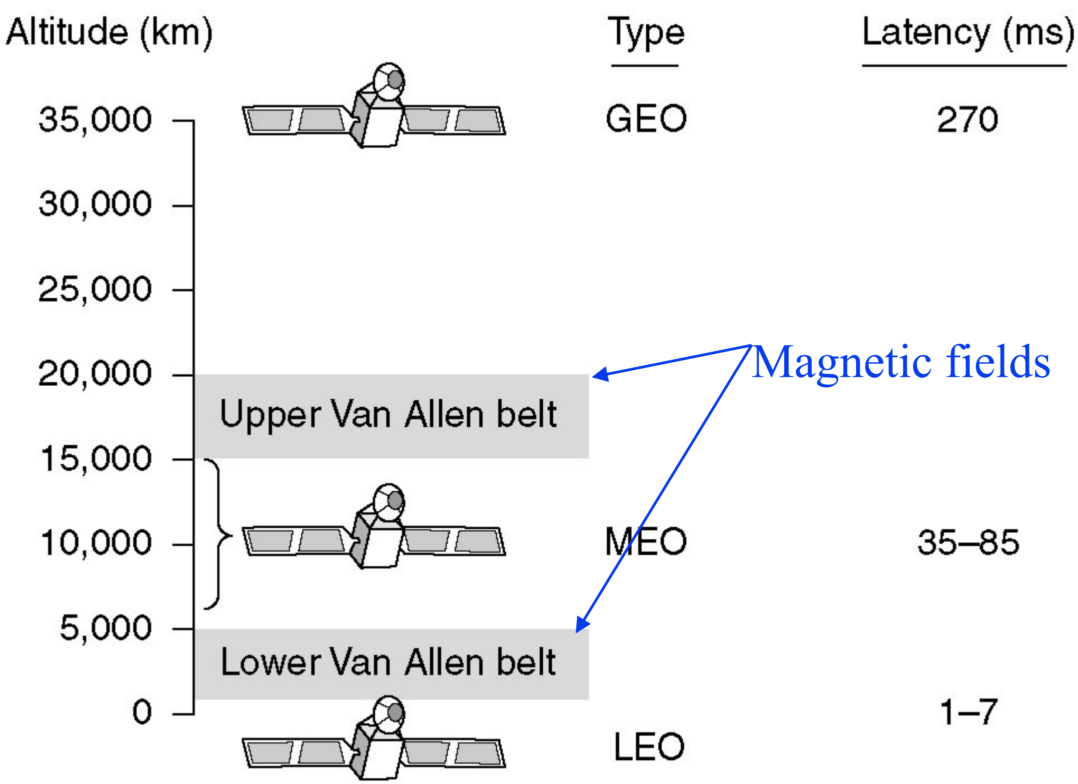
Wireless Transmission

- The Electromagnetic Spectrum
- Radio Transmission
- Microwave Transmission
- Infrared and Millimeter Waves
- Lightwave Transmission

Communication Satellites

- Geostationary Satellites
- Medium-Earth Orbit Satellites
- Low-Earth Orbit Satellites
- Satellites versus Fiber

Communication Satellites



Communication satellites and some of their properties, including altitude above the earth, round-trip delay time and number of satellites needed for global coverage.

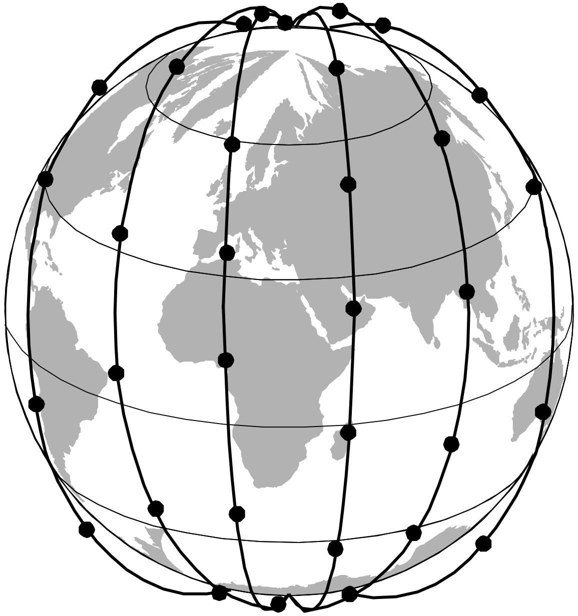
Why three regions for satellite placements?

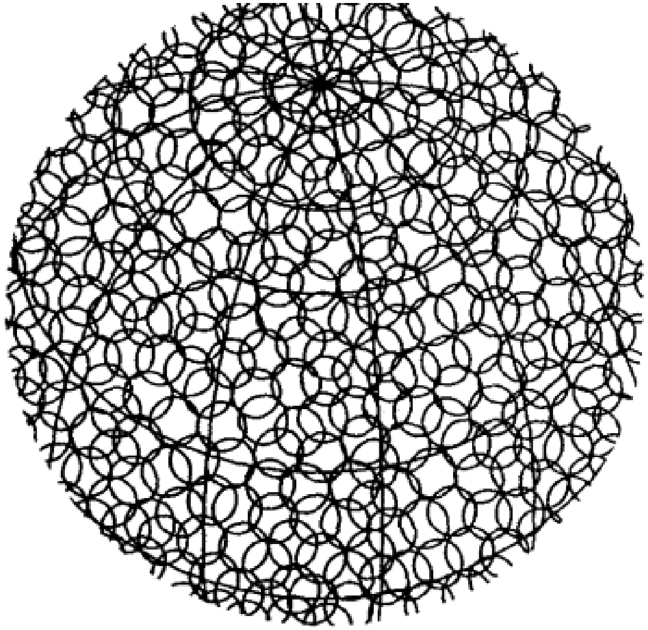
Where are the 24 GPS satellites?

Low-Earth Orbit Satellites Iridium

- Iridium: to provide worldwide telecommunication service using hand-held devices that communicate directly with the (66) Iridium satellites.

What are their behaviors at low-earth orbit?


(a)


(b)

(a) The Iridium satellites from six necklaces around the earth.

(b) 1628 moving cells cover the earth.

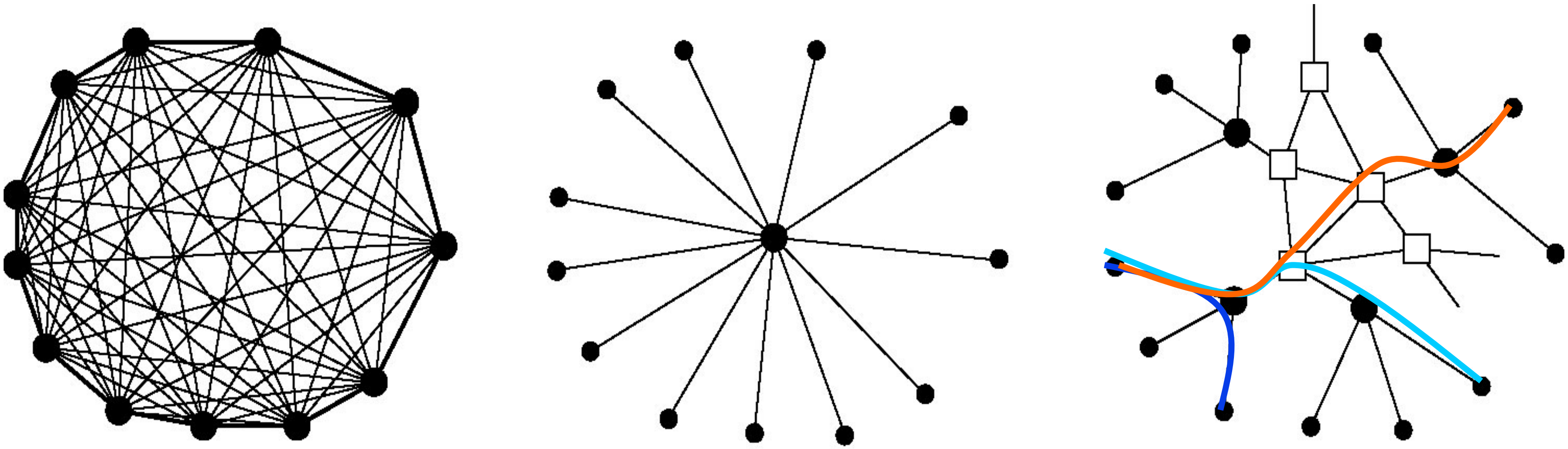
Why Iridium lost business?

Public Switched Telephone System

- Structure of the Telephone System
- The Politics of Telephones
- The Local Loop: Modems, ADSL and Wireless
- Trunks and Multiplexing
- Switching

Structure of the Telephone System

- PSTN (Public Switched Telephone Network):
  - voice or data (56 kbps vs. 1 Gbps)



(a) (b) (c)

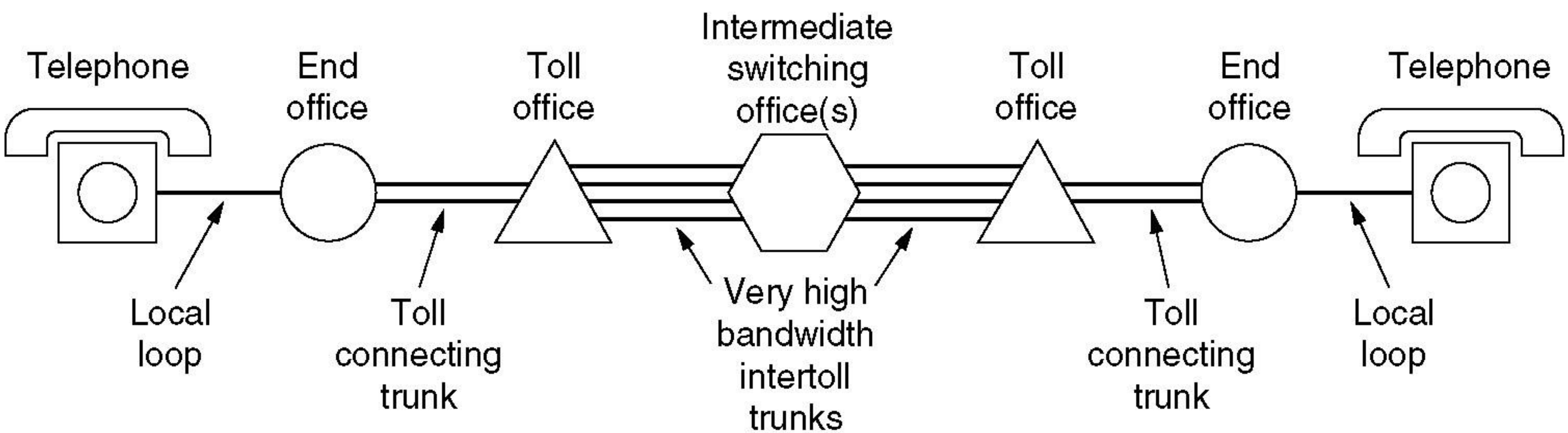
(a) Fully-interconnected network.

(b) Centralized switch.

(c) Two-level hierarchy.

Structure of the Telephone System (2)

- Three major components in the telephone system
  - Local loops ("last mile", analog - > digital)
  - Trunks (Multiplexing)
  - Switching offices



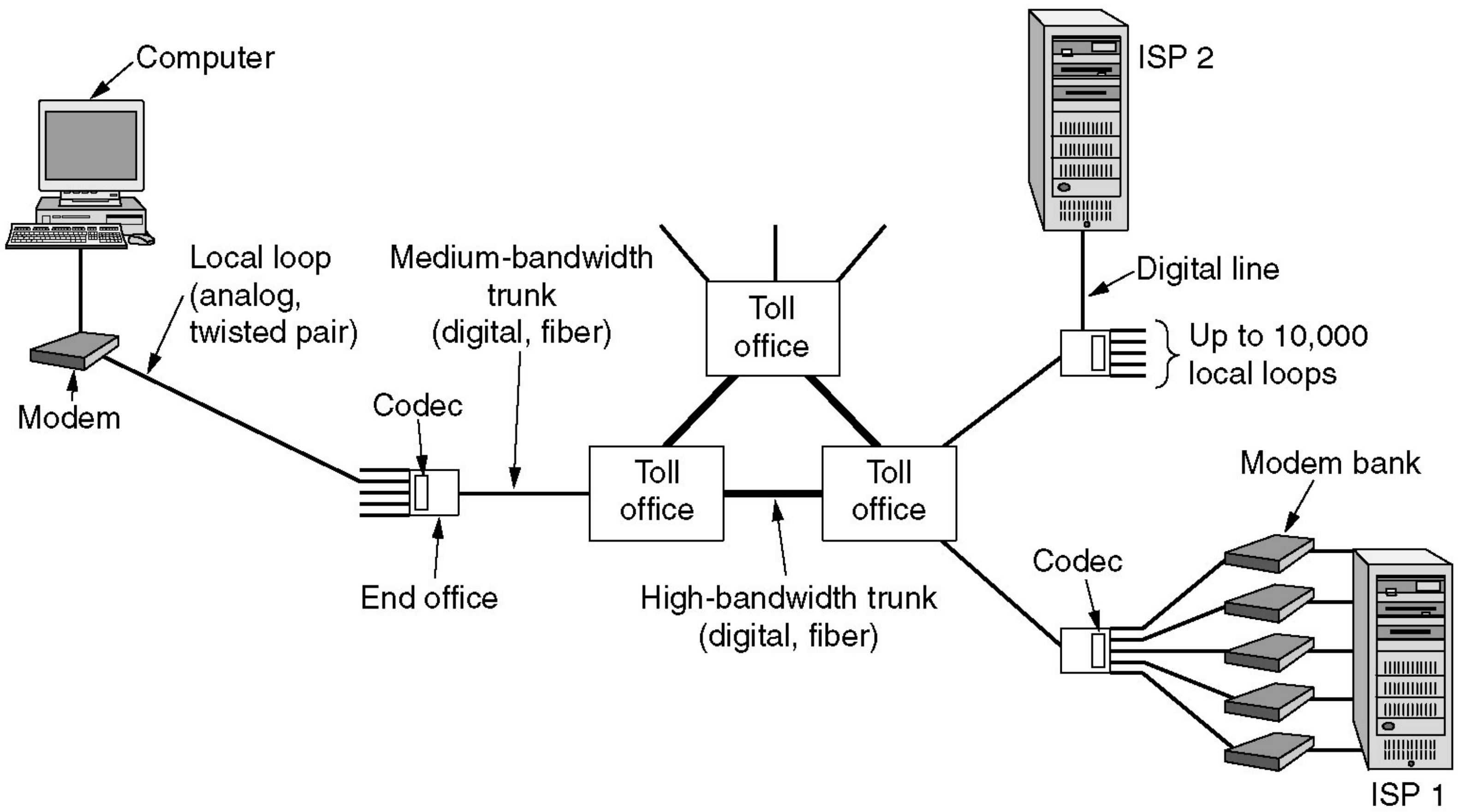
A typical circuit route for a medium-distance call.

Major Components of the Telephone System

- Local loops
  - Analog twisted pairs going to houses and businesses
- Trunks
  - Digital fiber optics connecting the switching offices
- Switching offices
  - Where calls are moved from one trunk to another

The Local Loop: Modems, ADSL, and Wireless

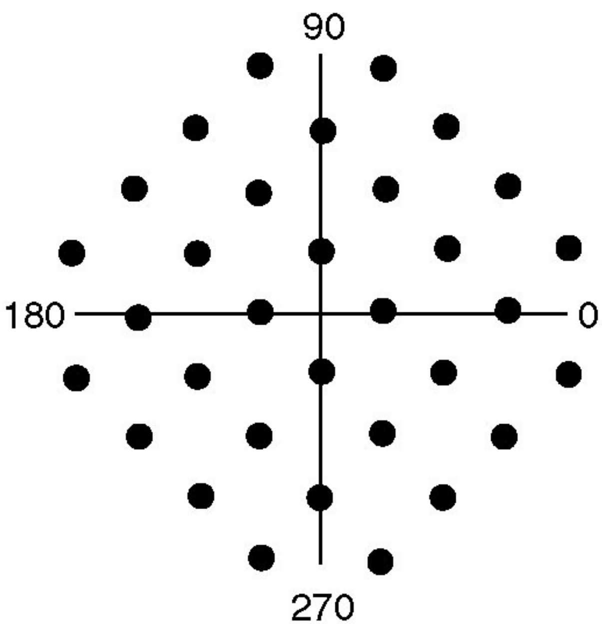
The use of both analog and digital transmissions for a computer to computer call. Conversion is done by the modems and codecs.



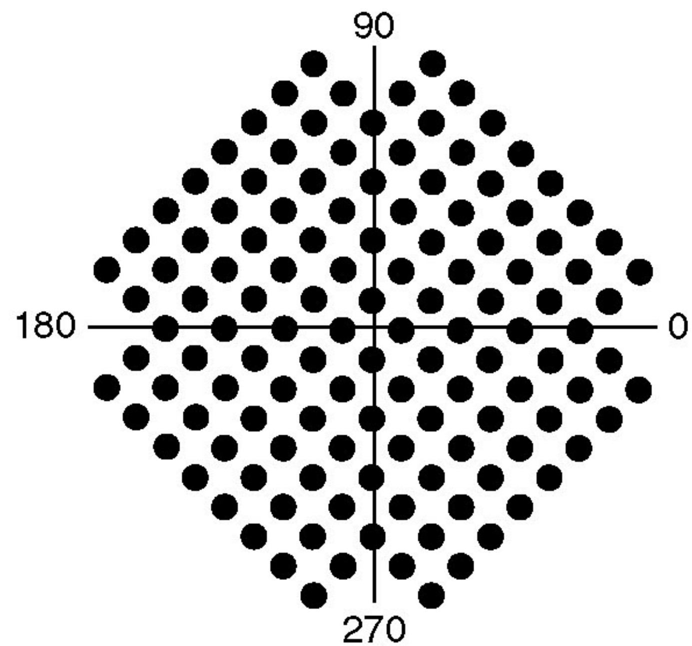
How many active connections can be handled by ISP1?

Modems

- Today's modem is called V.90
  - The baud rate: 8000 samples (symbols) /sec
  - The sample (symbol) size: 8 bits (7 data + 1 control in US)



(b)



(c)

TCM: (a) V.32 for 9600 bps.

(b) V.32 bits for 14,400 bps.

Simplex, Half Duplex, and Full Duplex

- Simplex connection
- Half-duplex connection
- Full duplex connection

Give an example per each type?

What does a modem belong to?

Topology Impact on Transmission Paths

A packet-switching network has a star topology with a central switch. It has n nodes. What are the best-, average-, and worst-case transmission paths in hops?

How about a bidirectional ring, and a fully interconnected packet-switching networks?

Answer:

star: 2, 2, 2 hops (best, average, worst)

ring: 1, N/4, N/2

full: 1, 1, 1

Trunks and Multiplexing

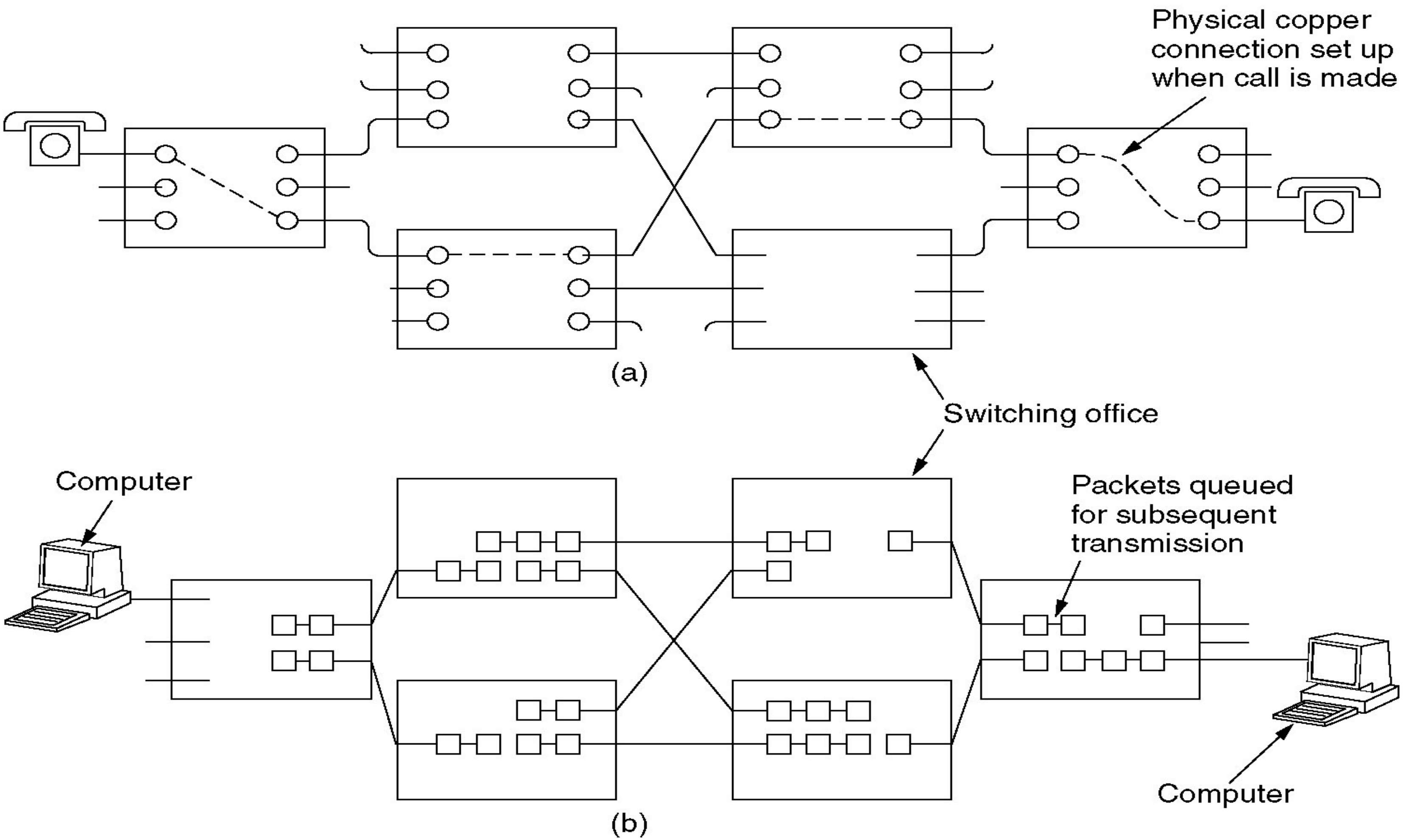
Multiplexing many conversations into a single physical trunk

- **FDM (Frequency Division Multiplexing)**
  - **The overall frequency spectrum is divided into frequency bands, with each user having exclusive possession of some band.**
- **TDM (Time Division Multiplexing)**
  - **The users take turns (Round-Robin), each one periodically getting the entire bandwidth for a little burst of time.**

Switching

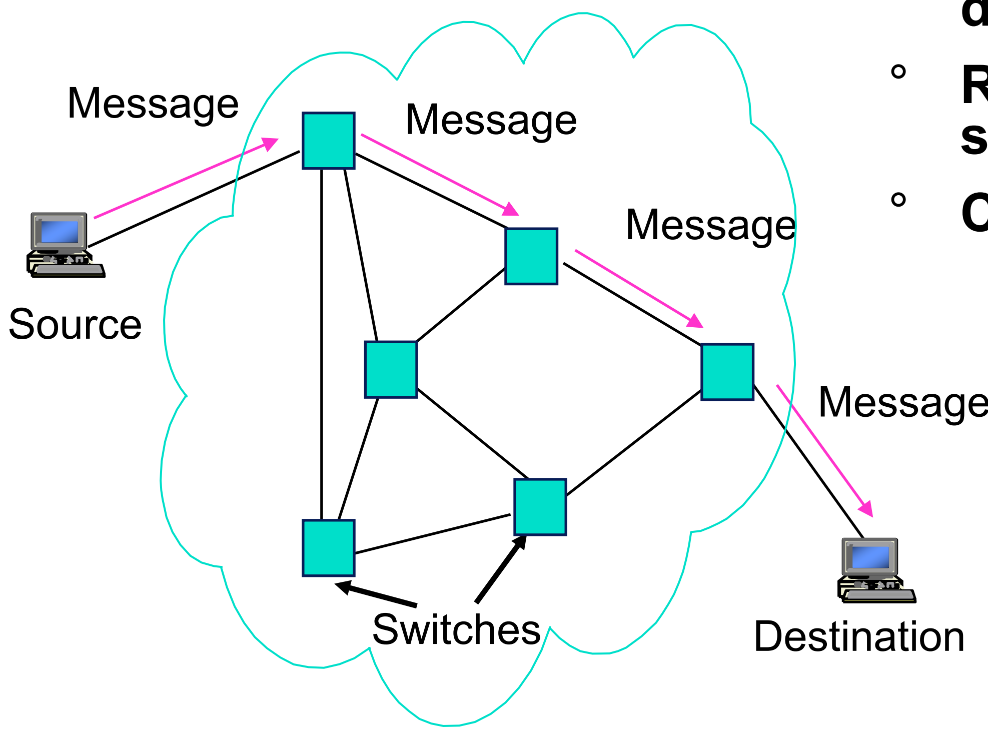
- Circuit Switching
- Message Switching
- Packet Switching

Circuit Switching vs. Packet Switching



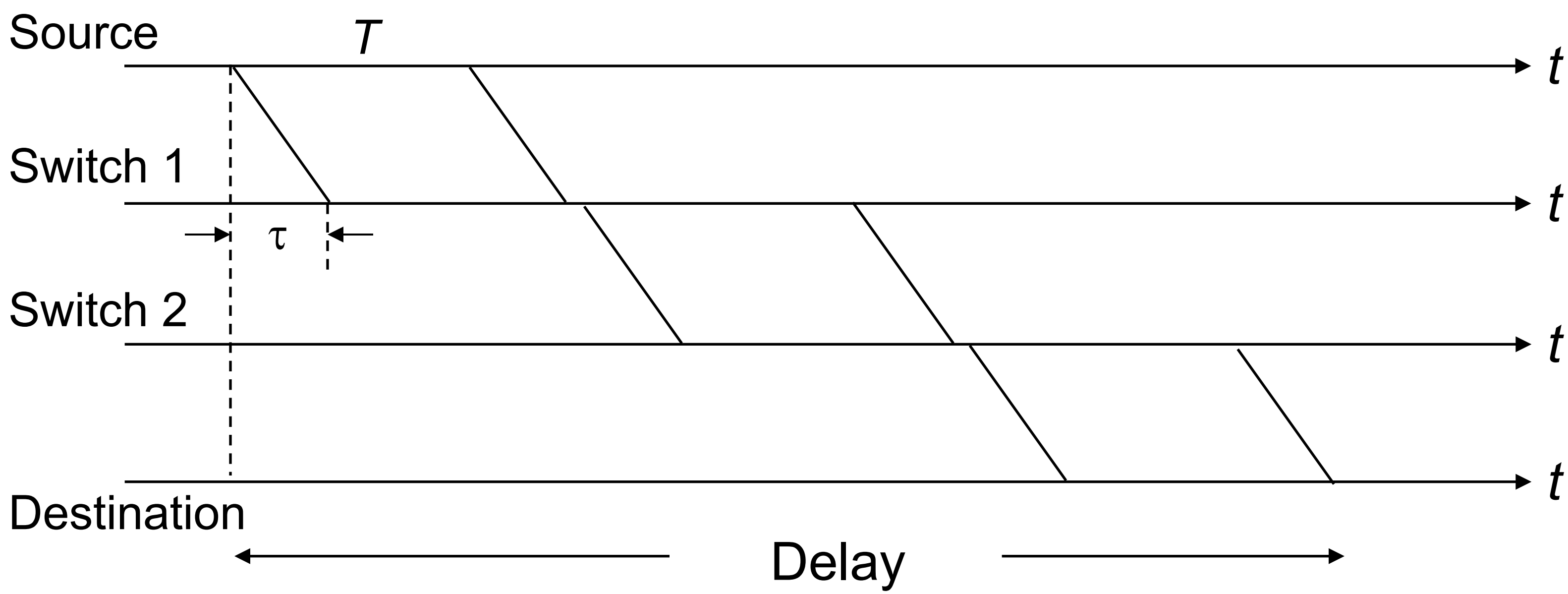
(a) Circuit switching. **(b) Packet switching.**

Message Switching



- Message switching invented for telegraphy
- Entire messages multiplexed onto shared lines, stored & forwarded
- Headers for source & destination addresses
- Routing at message switches
- Connectionless

Message Switching Delay

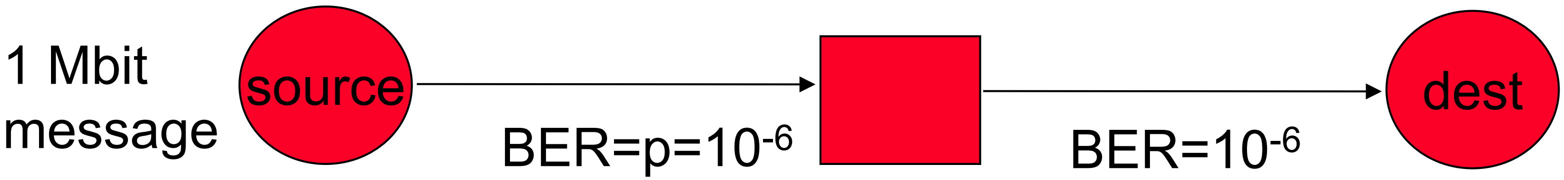


Minimum delay = $3\tau + 3T$

Additional queueing delays possible at each link

What are the major problems?

Long Messages vs. Packets



How many bits need to be transmitted to deliver message?

- Approach 1: send 1 Mbit message
- Probability message arrives correctly

$$P_c = (1-10^{-6})^{10^6} \approx e^{-10^6 10^{-6}} = e^{-1} \approx 1/3$$

- On average it takes about 3 transmissions/hop
- Total # bits transmitted ≈ 6 Mbits

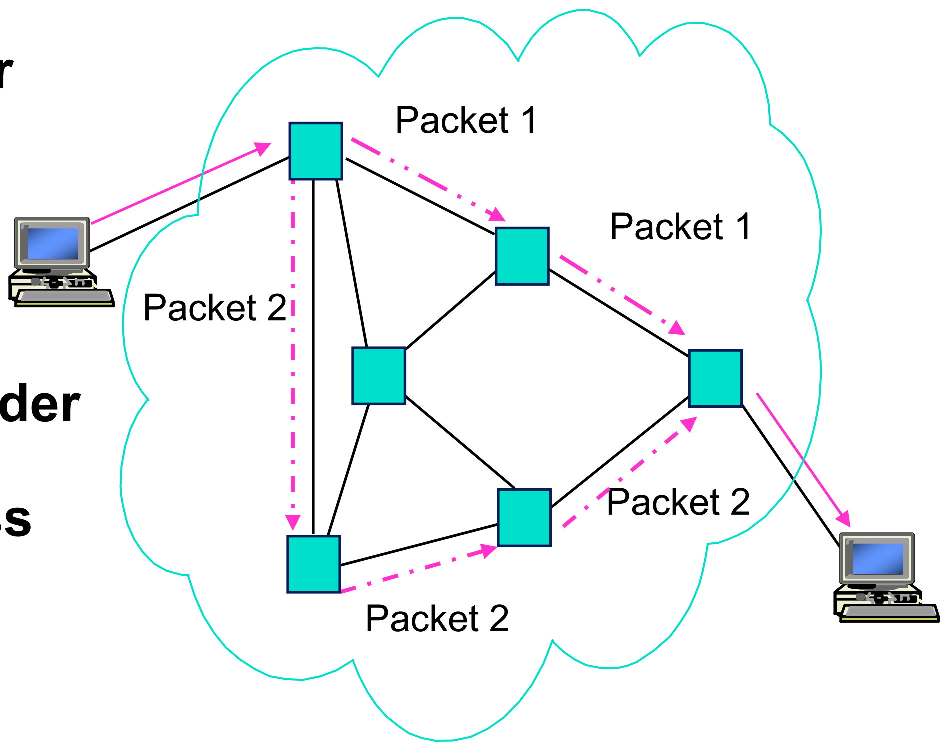
- Approach 2: send 10 100-kbit packets
- Probability packet arrives correctly

$$P_c' = (1-10^{-6})^{10^5} \approx e^{-10^5 10^{-6}} = e^{-0.1} \approx 0.9$$

- On average it takes about 1.1 transmissions/hop
- Total # bits transmitted ≈ 2.2 Mbits

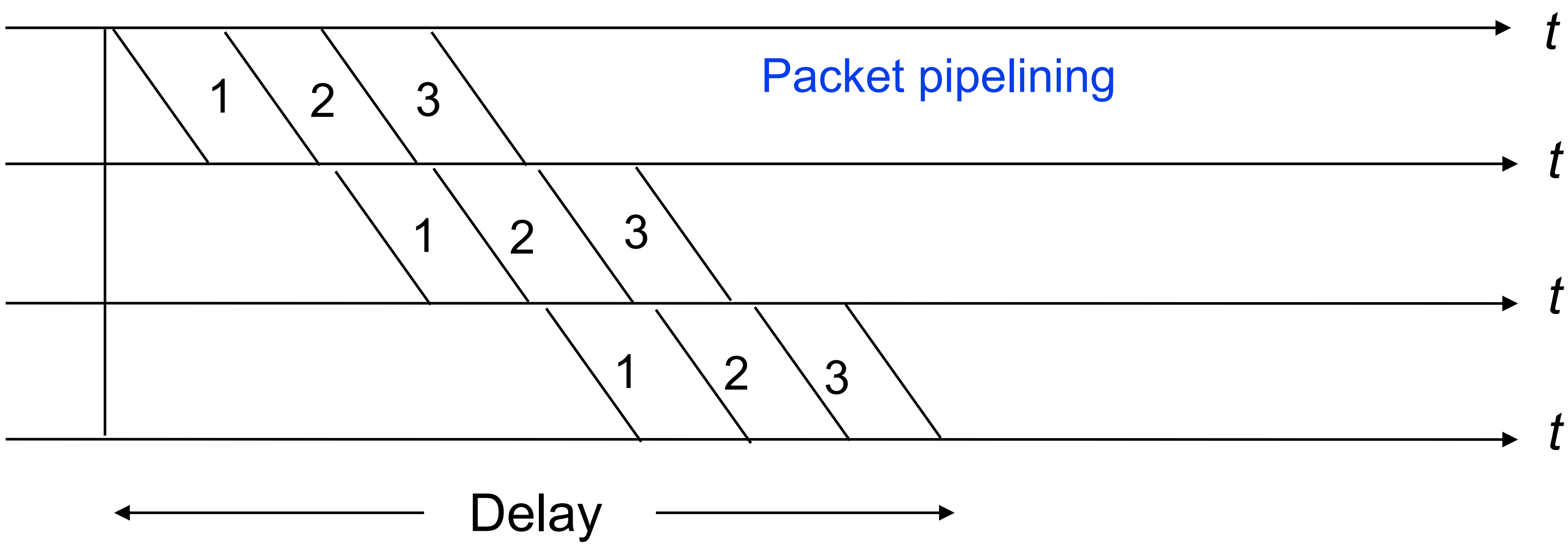
Packet Switching - Datagram

- Messages broken into smaller units (packets)
  - A packet has maximum size
- Source & destination addresses in packet header
- Connectionless, packets routed independently (datagram)
- Packet may arrive out of order
- Pipelining of packets across network can reduce delay, increase throughput
- Lower delay that message switching, suitable for interactive traffic



Packet Switching Delay

Assume three packets corresponding to one message traverse same path

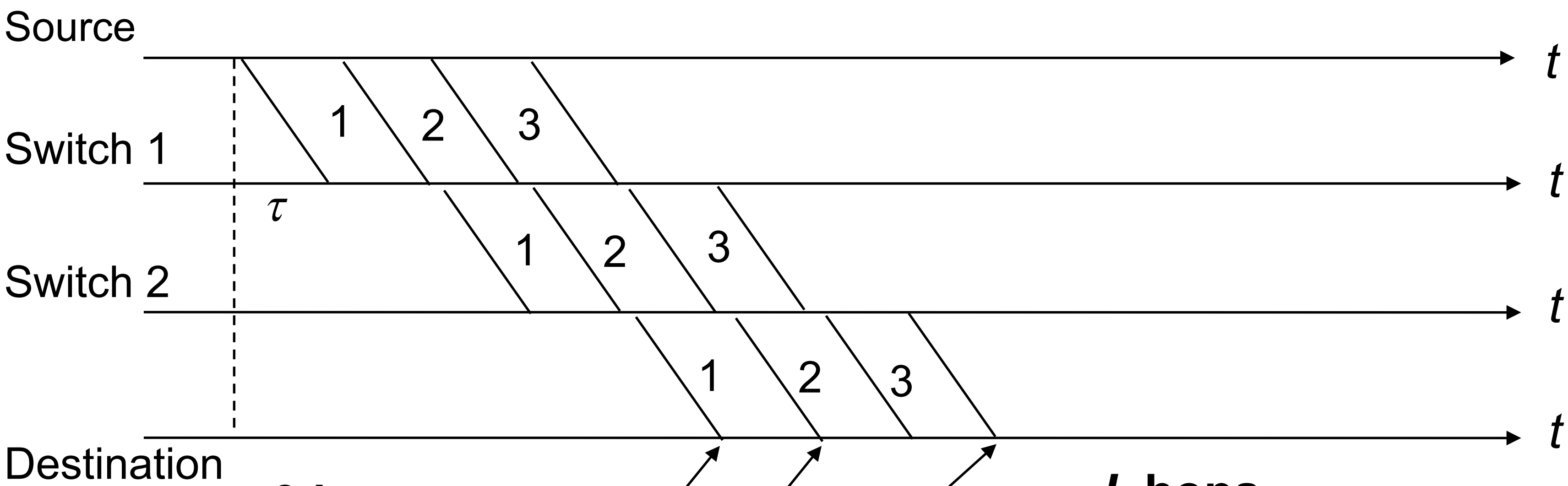


Minimum Delay = $3\tau + 5(T/3)$ (single path assumed)

Additional queueing delays possible at each link

Packet pipelining enables message to arrive sooner

Delay for k-Packet Messge over L Hops



3 hops

$3\tau + 2(T/3)$ first bit received

$3\tau + 3(T/3)$ first bit released

$3\tau + 5\,(T/3)$ last bit released

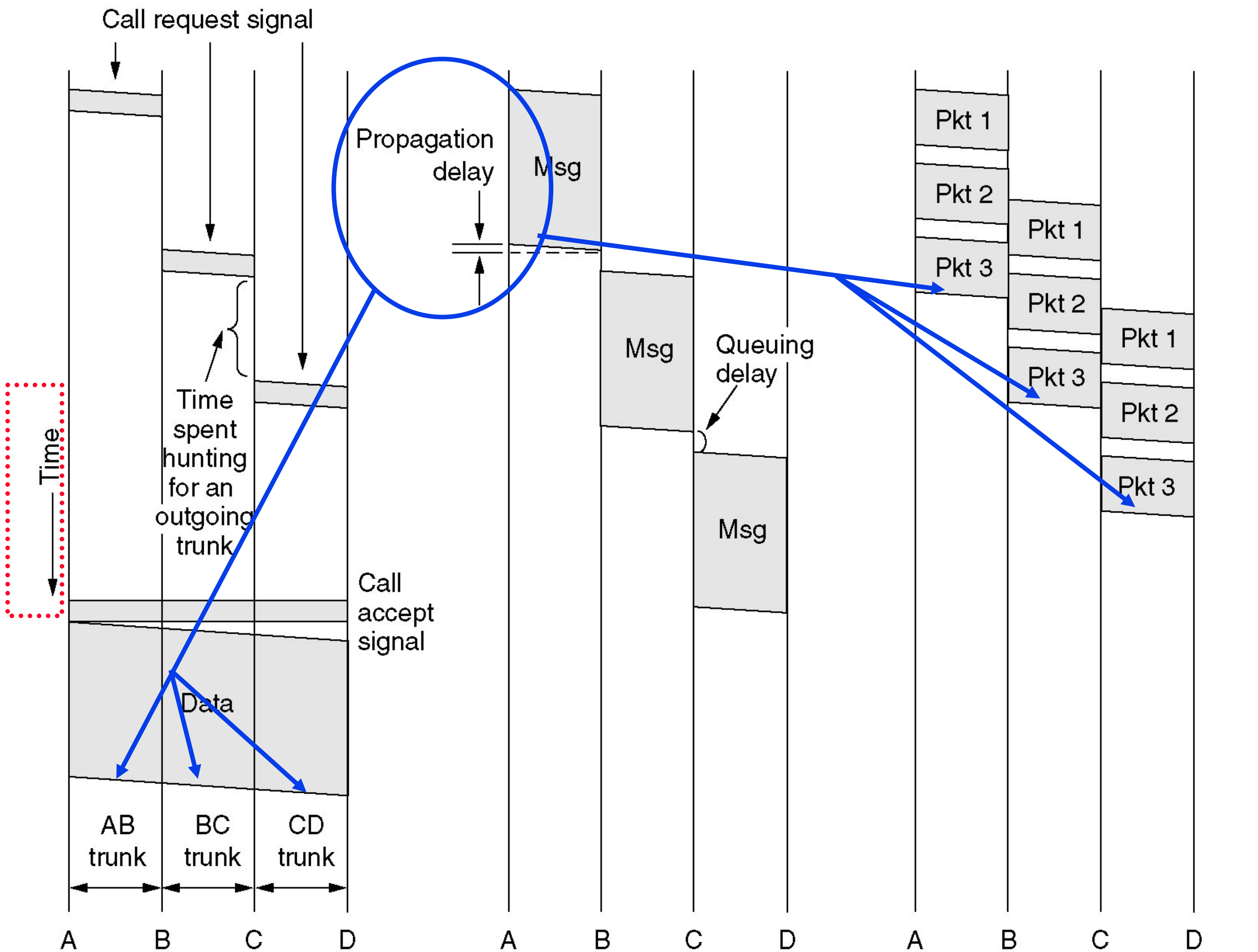
***L* hops**

$L\tau + (L\text{-}1)P$ first bit received

$L\tau + LP$ first bit released

$L\tau + LP + (k\text{-}1)P$ last bit released
or $L\tau + (L\text{-}1)P + kP$
where $T = k\,P$ *(what is P indeed?)*

Circuit, Message, Packet Switchings



(a) Circuit switching. (b) Message switching. (c) Packet switching

Switching Performance

- What is the delay of sending an ***X***-bit message over a ***L***-hop path in a circuit-switched network? The circuit setup time is ***s*** sec, the propagation delay is τ sec per hop, and the data rate is ***b*** bps.

- What is the delay of sending an ***X***-bit message over a ***L***-hop path in a lightly-loaded packet-switched network? The propagation delay per packet is τ sec per hop, the packet size is ***p*** bits, and the data rate is ***b*** bps.

$S+L^* \tau +X/b$

$X/b + (L-1)^*p/b$

Packet Switching

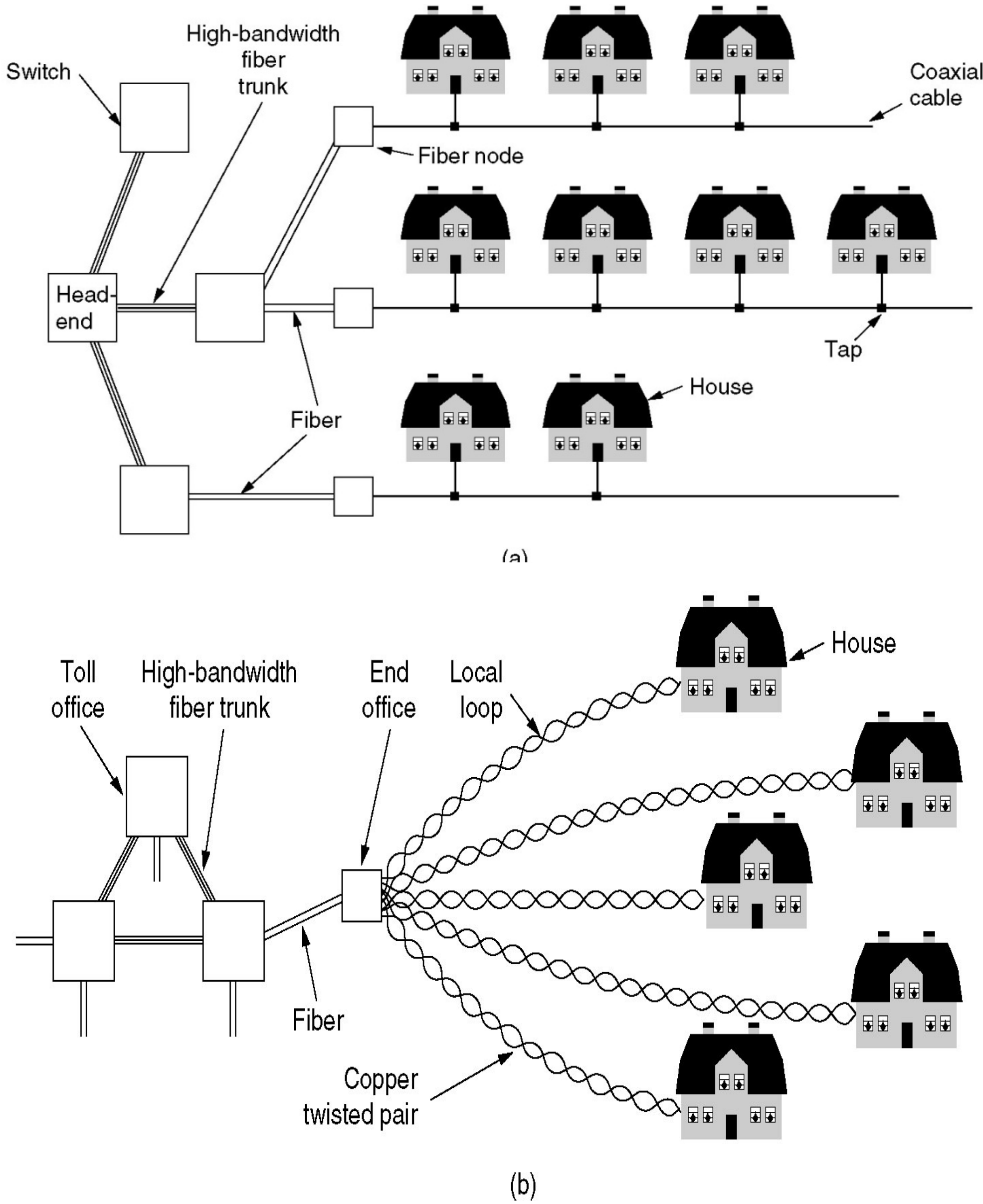
Item	Circuit-switched	Packet-switched
Call setup	Required	Not needed
Dedicated physical path	Yes	No
Each packet follows the same route	Yes	No
Packets arrive in order	Yes	No
Is a switch crash fatal	Yes	No
Bandwidth available	Fixed	Dynamic
When can congestion occur	At setup time	On every packet
Potentially wasted bandwidth	Yes	No
Store-and-forward transmission	No	Yes
Transparency	Yes	No
Charging	Per minute	Per packet

A comparison of circuit switched and packet-switched networks.

The Mobile Telephone System

- First-Generation Mobile Phones: Analog Voice
- Second-Generation Mobile Phones: Digital Voice
  - D-AMPS, GSM, CDMA
- Third-Generation Mobile Phones: Digital Voice and Data

Internet over Cable



What are two major challenges facing the cable operators if they want to get into the Internet/telephone business?

How the cable industry tackle the sharing problem?

Is there bandwidth competition between you and your neighbors if you use DSL?

Cable or DSL, which faster?

Reading and Homework

- Chapter 2 of the textbook.
  - Reading follows the lecturing – some concepts not covered in the lectures and their readings are optional therefore.

- Homework